

CSS Network Design

Jakob Sverre Alexandersen

GRA4154 Supply Chain Optimization with Mathematical Programming

Lecture 5 (guest)

February 25, 2026

Contents

1	Intro	2
2	Model	2

1 Intro

GreenCarbon (GC) is planning to build carbon capture and storage (CCS) facilities to capture and store CO₂ emissions from industrial sites. The plan is to build CO₂ capture facilities near industrial emission sources in Antwerp, Milan, Frankfurt, and Bilbao. GC has identified three potential locations for CO₂ storage facilities near Rotterdam, Hamburg, Marseille, and Copenhagen. Captured CO₂ emissions will be transported and stored underground in these storage facilities. The transport cost is 0.04 EUR per ton CO₂ per kilometer.

Table below demonstrates the distances (in km) between industrial emission sources and the potential CCS facilities

Source	Rotterdam	Hamburg	Marseille	Copenhagen
Antwerp	110	460	1050	900
Milan	990	1100	480	1300
Frankfurt	440	500	750	650
Bilbao	1200	1400	780	1750

The following table lists the fixed annual operating costs, maximum annual storage capacities, and variable storage costs at each potential facility

Facility	Fixed annual cost (EUR)	Storage Capacity	Variable storage cost
Rotterdam	2,000,000	250,000	12
Hamburg	1,800,000	200,000	11
Marseille	1,900,000	230,000	13
Copenhagen	1,400,000	150,000	10

Table below shows annual CO₂ emissions available t for capture from each industrial source:

Source	Annual Emissions
Antwerp	150,000
Milan	130,000
Frankfurt	140,000
Bilbao	120,000

2 Model

Sets

- $\mathcal{I} = \{\text{Antwerp, Milan, Frankfurt, Bilbao}\}$: emission sources
- $\mathcal{J} = \{\text{Rotterdam, Hamburg, Marseille, Copenhagen}\}$: storage facilities

Parameters

- d_{ij} : distance (km) from source i to facility j
- $t = 0.04$: transport cost (EUR/ton/km)
- f_j : fixed annual operating cost at facility j
- s_j : maximum storage capacity at facility j
- v_j : variable storage cost at facility j
- e_i : annual CO₂ emissions available for capture at source i

Decision variables

$$x_{ij} \geq 0 \quad \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (1)$$

$$y_j \in \{0, 1\} \quad \forall j \in \mathcal{J} \quad (2)$$

1. Tons of CO₂ transported from source i to facility j
2. 1 if facility j is opened, 0 otherwise

Objective function

$$\min Z = \sum_{j \in \mathcal{J}} f_j y_j + \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} (td_{ij} + v_j) x_{ij}$$

s.t.

$$\sum_{j \in \mathcal{J}} x_{ij} = e_i \quad \forall i \in \mathcal{I}$$

$$\sum_{i \in \mathcal{I}} x_{ij} \leq s_j y_j \quad \forall j \in \mathcal{J}$$

$$x_{ij} \geq 0 \quad \forall i \in \mathcal{I}, j \in \mathcal{J}$$

$$y_j \in \{0, 1\} \quad \forall j \in \mathcal{J}$$